GLV table rule and cycle escape above 40 bits (42-50)

Protocol: `PROTOCOL.md`, frozen in commit dbf722c (PR #1090) before any data on seed 20261031 existed; `sha256sum -c PROTOCOL.sha256` passes. Library: main `fb70171` plus `force-r.patch` (measurement flag only). Chart and analysis (`chart.py`) were written after the freeze; `analyze.py` is the frozen file. Counts of `group_ops` only; **no wall time**.

Run

- Arms: main (r = 32 at every size here), table 64, table 128, cycle escape off; controls r32 (identity) and r8 (positive) at 40 and 42 bits.
- Grid: one target per curve; curves per family 800/650/450/300/170/85 at 40/42/44/46/48/50 bits. 2 processes, `nice -n 10`, about 1.5 h each.
- Raw rows: `rows_<arm>_<part>.jsonl.gz`.

Invariants (all pass)

All answers correct in every arm and in the rho reference; every arm has the frozen grid with main's curves and targets; main's rule picks r = 32 for every curve; identity control: forced r = 32 gives identical `group_ops` on all 2,900 rows with 0 differing. Positive control passes: table 8 costs **1.203 [1.171, 1.235]** times main's operations at 40-42 bits.

Results

Ratios of curve_solve group operations to main on the same targets; intervals are 99.861% (Bonferroni over 36); ✓ fewer ops than main, × more, unmarked not resolved.

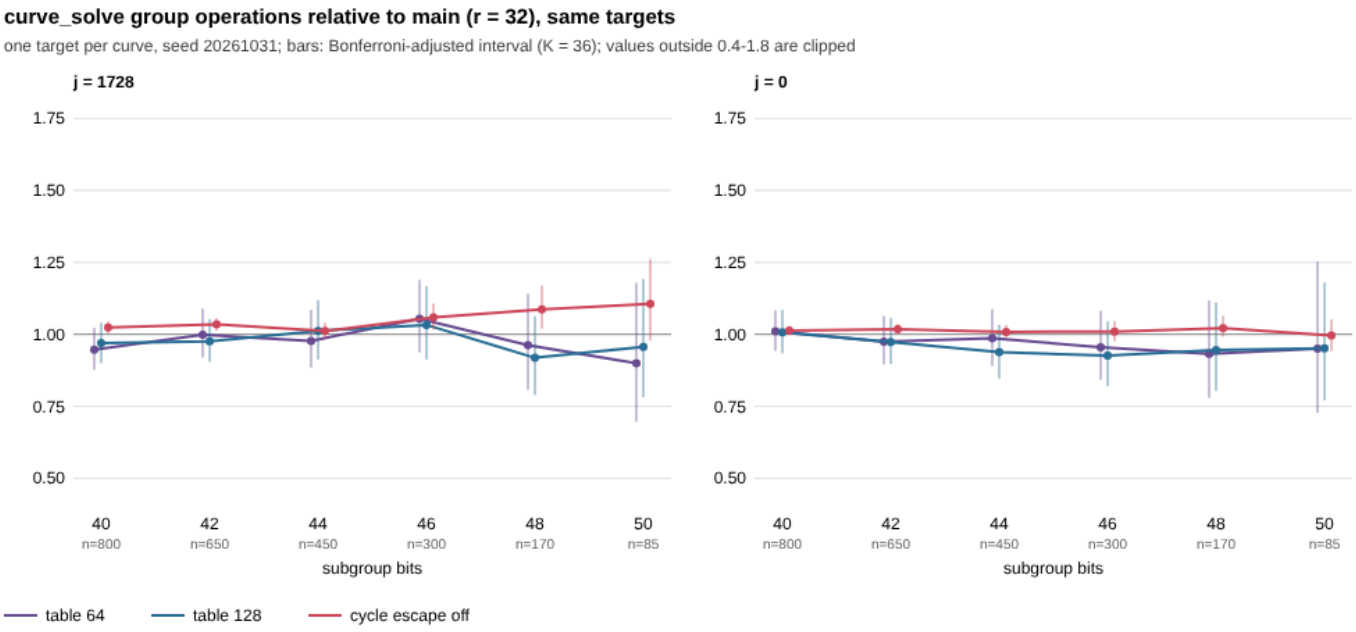
family	bits	main S	rho S	main/rho	r64/main	r128/main	no-escape/main
j1728	40	1.500	0.975	0.770	0.947 [0.878, 1.022]	0.969 [0.900, 1.041]	1.024 [1.002, 1.045] ×
j1728	42	1.434	0.939	0.764	0.998 [0.920, 1.090]	0.975 [0.905, 1.053]	1.035 [1.013, 1.057] ×
j1728	44	1.451	0.948	0.765	0.977 [0.885, 1.086]	1.012 [0.912, 1.119]	1.011 [0.984, 1.041]
j1728	46	1.397	0.924	0.756	1.054 [0.937, 1.189]	1.033 [0.912, 1.167]	1.059 [1.016, 1.107] ×
j1728	48	1.419	0.933	0.762	0.962 [0.808, 1.141]	0.918 [0.789, 1.063]	1.087 [1.020, 1.170] ×
j1728	50	1.486	1.024	0.722	0.900 [0.696, 1.179]	0.957 [0.782, 1.193]	1.106 [0.979, 1.262]
j0	40	1.475	0.987	0.611	1.011 [0.943, 1.083]	1.008 [0.934, 1.084]	1.012 [0.998, 1.029]
j0	42	1.452	0.968	0.610	0.975 [0.896, 1.064]	0.974 [0.897, 1.057]	1.018 [1.003, 1.036] ×
j0	44	1.492	0.917	0.666	0.987 [0.890, 1.088]	0.938 [0.846, 1.033]	1.009 [0.987, 1.032]
j0	46	1.472	0.941	0.640	0.955 [0.843, 1.082]	0.926 [0.820, 1.046]	1.009 [0.976, 1.047]
j0	48	1.423	0.917	0.636	0.932 [0.780, 1.118]	0.946 [0.803, 1.110]	1.022 [0.992, 1.065]
j0	50	1.363	0.934	0.599	0.950 [0.728, 1.253]	0.952 [0.771, 1.180]	0.996 [0.942, 1.052]

Positive control (table 8 vs main, 40 and 42 bits, 95% intervals):

- r8_vs_main_pooled_40_42: 1.203 [1.171, 1.235] pass=True
- r8:j1728:40: 1.180 [1.125, 1.239]
- r8:j1728:42: 1.281 [1.217, 1.349]
- r8:j0:40: 1.138 [1.087, 1.191]
- r8:j0:42: 1.166 [1.109, 1.228]

Descriptive pooled geometric mean of cell ratios, 42-50 bits (95%):

- r64:j1728: 0.977 [0.934, 1.022]
- r64:j0: 0.959 [0.917, 1.004]
- r64:both: 0.968 [0.936, 0.999]
- r128:j1728: 0.978 [0.939, 1.017]
- r128:j0: 0.947 [0.912, 0.986]
- r128:both: 0.962 [0.937, 0.991]
- noesc:j1728: 1.059 [1.039, 1.081]
- noesc:j0: 1.011 [1.001, 1.020]
- noesc:both: 1.035 [1.023, 1.046]



Observations

- **No larger table is resolved as better in any of the 12 cells.** All r64 and r128 adjusted intervals contain 1. Cell point ratios run 0.90-1.05 for r64 and 0.92-1.03 for r128, mostly below 1 at 46-50 bits, but the intervals are wide (power limit stated in the protocol: about +-12% at 42 bits, +-35% at 50 bits).
- **Descriptive, not a decision:** the pooled geometric mean over 42-50 bits is 0.968 [0.936, 0.999] (r64) and 0.962 [0.937, 0.991] (r128) for both families; j = 0 alone 0.959 / 0.947. This is a hint of a few percent, outside the Bonferroni decision rule, with intervals touching 1. It is a hypothesis for a held-out round, not a finding.
- **The escape still earns operations above 40 bits.** Switching it off costs 1.035 [1.023, 1.046] pooled over 42-50 bits; for j = 1728 it is resolved as more ops at 40, 42, 46 and 48 bits (up to 1.087 at 48), for j = 0 only at 42 bits (1.018). The earlier "gain vanishes above about 36 bits" does not hold for j = 1728 up to 48 bits in group operations. Caveat: the escape's doubling is an affine operation outside the batched inversion and is not priced here; whether the net wall effect is positive needs a Tier 1 run and the field-op accounting of the escape.
- **Distance to matched rho (group operations, unpriced):** curve_solve/rho is 0.72-0.77 for j = 1728 and 0.60-0.67 for j = 0 at every size, with no trend in the gap from 40 to 50 bits. S for curve_solve is about 1.36-1.50 (per sqrt(n/m)); rho S about 0.92-1.02 (per sqrt(n)). No method ranking is claimed (AGENTS.md 5a band not applied).

Limits

Single seed, one target per curve, 85 targets at 50 bits. Effects under roughly 5-10% per cell are not excluded. Operation counts, not time.

Proposed next step

A held-out confirmation round on a new seed, frozen first, comparing r = 64 or 128 with r = 32 pooled over 42-46 bits, with the sample size set from a power calculation for a 3% effect. Any wall-time claim needs a Tier 1 run, which is the user's decision. No code-change PR is opened.